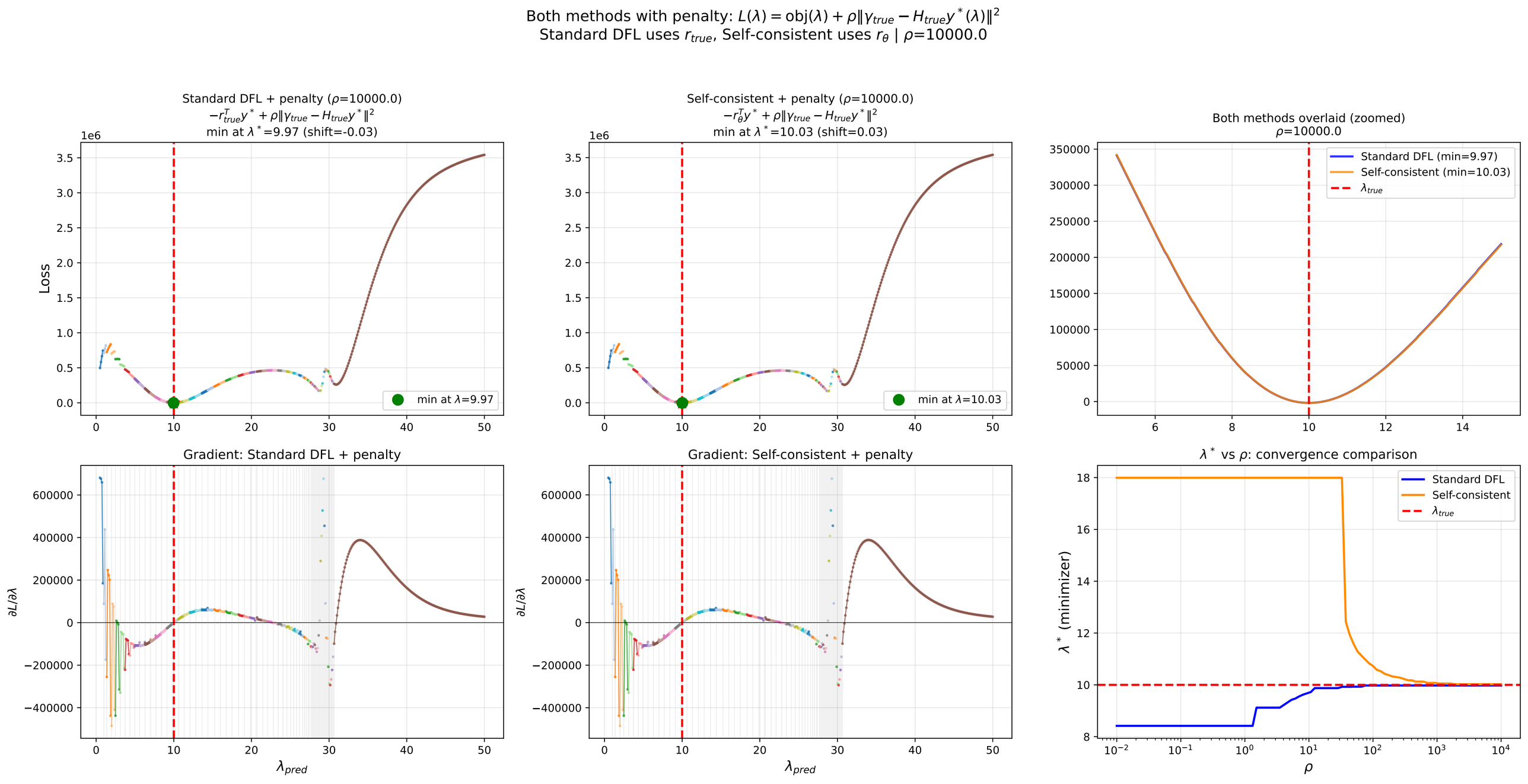

Both methods with penalty: $L(\lambda) = \text{obj}(\lambda) + \rho\|\gamma_{true} - H_{true}y^*(\lambda)\|^2$
Standard DFL uses r_{true}, Self-consistent uses r_θ | ρ=10000.0
Standard DFL + penalty (ρ=10000.0)
$-r_{true}^T y^* + \rho\|\gamma_{true} - H_{true}y^*\|^2$
min at λ^*=9.97 (shift=-0.03)
1e6
Loss
min at λ=9.97
Self-consistent + penalty (ρ=10000.0)
$-r_\theta^T y^* + \rho\|\gamma_{true} - H_{true}y^*\|^2$
min at λ^*=10.03 (shift=0.03)
min at λ=10.03
Both methods overlaid (zoomed)
ρ=10000.0
Standard DFL (min=9.97)
Self-consistent (min=10.03)
λ_{true}
Gradient: Standard DFL + penalty
Gradient: Self-consistent + penalty
$\partial L/\partial\lambda$
λ_{pred}
λ^* vs ρ: convergence comparison
Standard DFL
Self-consistent
λ^* (minimizer)
ρ